

MANISH REDDY

AI Data Engineer

Databricks Data Engineer Professional | Databricks Generative AI Engineer Associate | PySpark | Azure | Kafka | RAG
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GitHub: <https://github.com/Reddy-datahub> • Vercel: [URL]

PROFESSIONAL SUMMARY

AI Data Engineer with 5+ years of experience designing and delivering scalable data platforms, ETL/ELT pipelines, real time streaming solutions and AI driven data workflows across telecom, digital banking and insurance. Strong expertise in Python, PySpark, Spark SQL, SQL, Kafka, Azure, PostgreSQL and MySQL with hands-on experience building reliable batch and streaming data pipelines. Experienced in applying machine learning for feature engineering, anomaly detection, fraud risk analysis and intelligent data validation along with developing GenAI data solutions using RAG, semantic search, vector retrieval and LLM ready knowledge pipelines. Skilled in integrating structured and unstructured data developing REST APIs and deploying production data and AI workloads using Docker, Kubernetes/AKS, Terraform, Jenkins and CI/CD.

CORE COMPETENCIES

Languages	Python, SQL
Data Engineering	PySpark, Spark SQL, ETL/ELT, Data Pipelines, Data Transformation, Data Quality, Data Validation, Data Reconciliation
Streaming & Databases	Apache Kafka, PostgreSQL, MySQL, Elasticsearch
AI/ML & GenAI	Machine Learning, Feature Engineering, Anomaly Detection, RAG, LLM, Vector Search, Semantic Search
Cloud & DevOps	Azure, AKS, Kubernetes, Docker, Terraform, Jenkins, Git, CI/CD
APIs & Integration	REST APIs, Data Services, Third-Party API Integration

PROFESSIONAL EXPERIENCE

Bell Canada

Toronto, ON

Data Engineer - AI & GenAI Platform

Jul 2024 – Present

Designed and developed scalable Azure data and AI pipelines using Python, PySpark, Spark SQL, Kafka and Azure data services to integrate customer, billing, order, service provisioning and support data for analytics and GenAI applications.

- Built enterprise ETL/ELT pipelines with PySpark and Spark SQL to ingest, cleanse, transform, enrich and curate structured and semi structured telecom data, creating trusted and AI ready datasets for downstream analytics and LLM applications.
- Engineered real time Kafka streaming pipelines to process customer, service, order and support events, enabling near real time data availability for analytics, intelligent customer support workflows and AI retrieval applications.
- Built data pipelines supporting Retrieval Augmented Generation by integrating knowledge articles, troubleshooting guides, service policies, product documentation and historical support cases into AI ready knowledge pipelines.
- Developed document ingestion and knowledge engineering pipelines using Python for document parsing, preprocessing, chunking, metadata enrichment, deduplication, indexing and incremental refreshes to support semantic retrieval and LLM applications.
- Integrated Elasticsearch vector search and semantic retrieval into GenAI data workflows enabling similarity-based retrieval of relevant enterprise knowledge and historical support information for grounded LLM applications.
- Developed Python based REST APIs and AI data services to expose customer, service, knowledge and retrieval data to AI applications, implementing validation, exception handling, logging and secure enterprise data access.
- Containerized and deployed data, search and AI workloads using Docker and Kubernetes/AKS while implementing CI/CD, Terraform-based infrastructure automation, Azure Monitor, data quality checks and operational monitoring for reliable production AI data services.

Key Skills & Technologies: Python, PySpark, Spark SQL, Kafka, Azure, Elasticsearch, RAG, LLM, REST APIs, Docker, Kubernetes/AKS, Terraform, Jenkins, Git.

CIBC

Toronto, ON

Data Engineer – AI/ML

May 2022 – Jun 2024

Designed ETL/ELT and event driven data pipelines using Python, PySpark, SQL and Kafka to support customer onboarding, account creation, transfers, bill payments and digital banking workflows.

- Built Python and PySpark data processing pipelines to cleanse, transform, enrich and prepare KYC, customer, account and transaction data for analytics and machine learning use cases.
- Engineered feature engineering and data enrichment workflows to generate customer, account and transaction level features supporting fraud risk analysis and anomaly detection models.

- Implemented machine learning based anomaly detection workflows to identify unusual transaction and account activity, generating risk indicators for downstream fraud and operational review.
- Developed ML-assisted KYC and document validation workflows to identify missing, inconsistent or potentially problematic customer and onboarding information for exception handling.
- Integrated third party identity, credit bureau and notification data into curated data layers with schema validation, data quality checks, error handling and retry mechanisms.
- Implemented data and machine learning validation controls covering missing values, duplicates, outliers, schema consistency and model-input quality to improve the reliability of downstream analytics and ML workflows.
- Optimized PostgreSQL schemas indexes and SQL queries for high volume banking data while supporting Docker based deployments and Jenkins CI/CD automation.

Key Skills & Technologies: Python, PySpark, Spark SQL, SQL, Kafka, PostgreSQL, Machine Learning, Feature Engineering, Anomaly Detection, Docker, Git.

Manulife

Vancouver, BC

Data Engineer

Jan 2021 – Mar 2022

Built Python and SQL based data pipelines supporting policy issuance, customer profiles, premium processing, renewals and claims status workflows.

- Developed ETL processes and data transformations to extract, cleanse, validate and load insurance policy and customer data into relational data stores.
- Created and optimized MySQL queries, stored procedures and database objects supporting policy administration, premium calculations, renewals and operational reporting.
- Implemented data validation and reconciliation checks for policy attributes, premium inputs, customer information and claims-status records to improve data accuracy.
- Developed REST based data services to expose curated policy and customer information for internal applications and downstream reporting workflows.
- Supported production data troubleshooting and quality improvements, resolving failed loads, data inconsistencies and processing issues across policy and premium workflows.

Key Skills & Technologies: Python, SQL, MySQL, ETL, REST APIs, Data Validation, Data Reconciliation, Git, JUnit.

EDUCATION

Bachelor of commerce , Computer science | Osmania university | hyderabad, India | 2019

CERTIFICATIONS

- **Databricks Certified Data Engineer Professional** | Issued by Databricks | 2026
- **Databricks Certified Generative AI Engineer Associate** | Issued by Databricks | 2026